

SCHEDULING ALGORITHM

1)

```
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++)
    {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

```
mahij@Mahija: ~  
mahij@Mahija:~$ nano fcfs.c  
mahij@Mahija:~$ gcc fcfs.c -o fcfs  
mahij@Mahija:~$ ./fcfs  
Enter number of processes: 3  
Enter burst time for each process:  
P1: 5  
P2: 3  
P3: 2  
  
Process BT      WT      TAT  
P1      5      0      5  
P2      3      5      8  
P3      2      8     10  
mahij@Mahija:~$ |
```

2)

```

GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort burst times
    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (bt[i] > bt[j])
            {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++)
    {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}

```

```
mahij@Mahija:~$ nano sjfsnonprem.c
mahij@Mahija:~$ gcc sjfsnonprem.c -o sjfsnonprem
mahij@Mahija:~$ ./sjfsnonprem
Enter number of processes: 3
Enter burst time:
P1: 5
P2: 2
P3: 3

Process BT      WT      TAT
P1      2      0      2
P2      3      2      5
P3      5      5     10
mahij@Mahija:~$ |
```

3)

```

GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    // Sort according to priority
    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (priority[i] > priority[j])
            {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++)
    {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t\t%d\t%d\n",
            i + 1, bt[i], priority[i], wt[i], tat[i]);
    }

    return 0;
}

```

```
mahij@Mahija:~$ nano priority.c
mahij@Mahija:~$ gcc priority.c -o priority
mahij@Mahija:~$ ./priority
Enter number of processes: 3
Enter burst time and priority:
P1 Burst Time: 5
P1 Priority: 3
P2 Burst Time: 2
P2 Priority: 1
P3 Burst Time: 3
P3 Priority: 2

Process BT      Priority      WT      TAT
P1      2        1          0        2
P2      3        2          2        5
P3      5        3          5       10
mahij@Mahija:~$ |
```

4)

```

GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);

        rem_bt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &quantum);

    // Round Robin
    while (1)
    {
        done = 1;

        for (i = 0; i < n; i++)
        {
            if (rem_bt[i] > 0)
            {
                done = 0;

                if (rem_bt[i] > quantum)
                {
                    time = time + quantum;
                    rem_bt[i] = rem_bt[i] - quantum;
                }
                else
                {
                    time = time + rem_bt[i];
                    wt[i] = time - bt[i];
                    rem_bt[i] = 0;
                }
            }
        }

        if (done == 1)
            break;
    }

    // Turnaround time
    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}

```

```
mahij@Mahija:~$ nano roundrobin.c
mahij@Mahija:~$ gcc roundrobin.c -o roundrobin
mahij@Mahija:~$ ./roundrobin
Enter number of processes: 3
Enter burst time:
P1: 5
P2: 3
P3: 2
Enter time quantum: 2

Process BT      WT      TAT
P1      5      5      10
P2      3      6      9
P3      2      4      6
mahij@Mahija:~$ |
```